

7. Public Hackathon Explore Page 🌐

Ye bhi important hai.

Participant ko sirf apne college ke hackathons nahi dikhne chahiye.

Usse platform ke **saare published public hackathons** explore karne milenge.

Example:

Explore Hackathons

🔍 Search hackathons...

Filters:

[Online]

[Offline]

[Hybrid]

Category:

[AI/ML]

[Web]

[Cyber Security]

[Open Innovation]

Status:

[Registration Open]

[Upcoming]

Card:

Hackathon Banner
TechSprint 2026
Online • 3,245 Participants
Registration Open
[View Details]

Participant apni choice se kisi bhi public hackathon ko open karke register karega.

8. Hackathon Details Page

Ye page **sabse important public page** me se ek hoga.

Isme:

Banner

Hackathon Name

Organization

Description



Important Dates



Team Size



Prizes



Rules



Submission Requirements



Judging Criteria



FAQs

[Register Now]

Aur agar registration close ho chuki hai:

Registration Closed

System automatically button change karega.

Golden Rule ★

The interface should show users what they need, not everything the system knows.

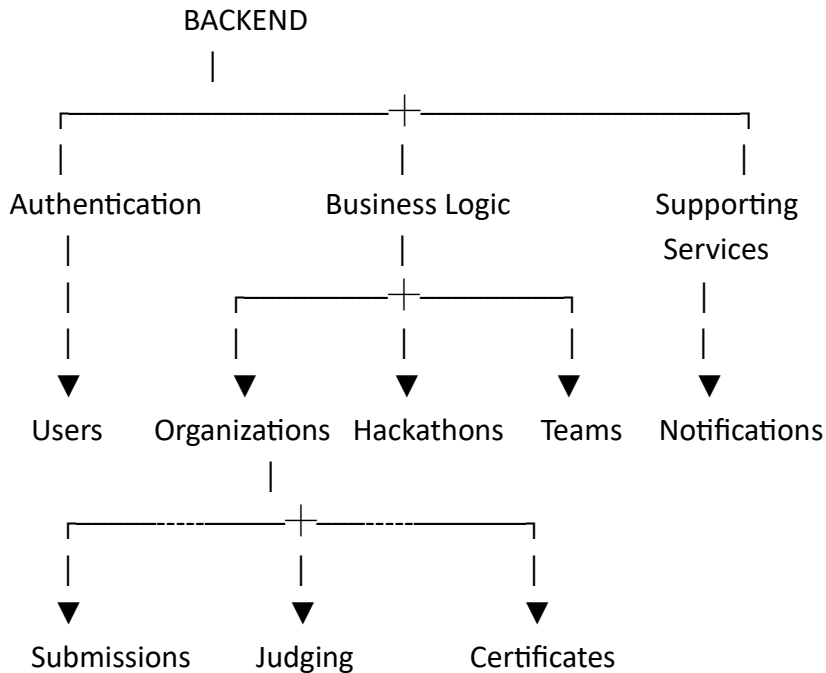
Participant ko database ke 50 tables ki information nahi chahiye. 😊

Usko bas woh information chahiye jo uske current task ke liye relevant hai.

Chapter 38: Backend Architecture.

Backend ka kaam business logic, security, data processing aur frontend ke requests ko handle karna hai. Hamare case me backend ko ek hi giant file nahi banana hai. Usko logical modules me divide karenge.

1. Overall Backend Structure



2. Authentication Module 🗝️

Iska kaam:

- Signup
- Login
- Logout
- Password reset
- Google authentication
- Token/session handling
- Authentication checks

Example:

POST /auth/signup

POST /auth/login

POST /auth/logout

POST /auth/forgot-password

3. User Module

Iska kaam:

- Profile
- User information
- Skills
- Profile photo
- User roles
- User settings

Example:

GET /users/me

PATCH /users/me

4. Organization Module

Iska kaam:

- Create organization
- Organization profile
- Add organizers
- Remove organizers
- Transfer ownership
- Organization permissions

Example:

POST /organizations

GET /organizations/{id}

PATCH /organizations/{id}

5. Hackathon Module

Ye hamare platform ka **central module** hoga.

Isme:

- Create hackathon
- Draft
- Configure
- Publish
- Update
- Registration
- Status management
- Rules
- Dates
- Team settings

Example:

POST /hackathons

GET /hackathons

GET /hackathons/{id}

PATCH /hackathons/{id}

POST /hackathons/{id}/publish

6. Team Module

Iska kaam:

- Create team
- Invite member
- Accept invitation
- Join team
- Team settings
- Team freeze
- Team members

Example:

POST /teams

POST /teams/{id}/invite

POST /teams/{id}/join

GET /teams/{id}

7. Submission Module

Iska kaam:

- Upload project
- Save file references
- GitHub URL
- Demo URL
- Submission version
- Submit
- Lock after deadline

Example:

POST /submissions

PATCH /submissions/{id}

POST /submissions/{id}/submit

GET /submissions/{id}

8. Judging Module

Iska kaam:

- Add judges
- Assign projects
- Evaluation criteria

- Score
- Feedback
- Review flags
- Final evaluation

Example:

POST /judges

POST /assignments

POST /evaluations

PATCH /evaluations/{id}

9. Result Module 🏆

Iska kaam:

- Calculate final score
- Apply tie-break rules
- Generate rankings
- Publish results
- Leaderboard

Example:

GET /hackathons/{id}/results

POST /hackathons/{id}/publish-results

10. Certificate Module 📄

Iska kaam:

- Generate certificates
- Certificate ID
- QR verification
- Participation certificate
- Winner certificate
- Judge certificate

Example:

GET /certificates/{id}

GET /certificates/{id}/verify

11. Notification Module 🔔

Ye background me kaam karega.

Example:

Team invitation

↓

Notification Service

↓

Email + In-app notification

Ye hackathon deadline reminders bhi bhej sakta hai.

12. Analytics Module 📊

Organizer ko:

- Registrations
- Teams
- Submissions
- Completion rate
- Judging progress

dikhayega.

Super Admin ko platform-level analytics milengi.

Backend ka Golden Rule ★

Each module should have a clear responsibility.

Agar hackathon ka code 5,000 lines ki ek file me bhar diya, to project maintain karna mushkil ho jayega.

Ab ek important architecture concept

Suppose participant registration karta hai:

Participant



Frontend



POST /hackathons/123/register



Backend



Authentication



Hackathon Module



Check:

Registration open?



Team requirements?



Already registered?



Eligible?



Database



Success

Notice karo: **API endpoint sirf request receive karta hai.**

Actual rules/business logic backend ke appropriate module me honi chahiye.

Next Important Decision

Ab backend ko database se baat karni hai.

Do common approaches hain:

Option A: Backend ke andar SQL queries directly likho.

Backend → SQL → Database

Option B: Backend me ek **ORM (Object-Relational Mapping)** layer use karo.

Backend



ORM



Database

ORM developer ko database ke saath Python objects/models ke through kaam karne deta hai.

Since hum likely FastAPI + Python use karenge, **SQLAlchemy** jaise ORM ka option aa sakta hai.

Option A: Direct SQL

Maan lo tum restaurant gaye.

Tum chef ko bolte ho:

"2 onion, 1 tomato, 5g salt, medium flame, 8 minutes cook..."

Matlab tum har instruction khud de rahe ho.

Database me bhi aisa hi hota hai.

```
SELECT * FROM users;
```

```
INSERT INTO hackathons (...);
```

Yaani tum database se uski language (SQL) me baat karte ho.

Advantages

- ✓ Fast
- ✓ Full control

Disadvantages

- ✗ Bahut saari SQL manually likhni padti hai.
- ✗ Bade projects me maintain karna mushkil ho sakta hai.

Option B: ORM

Ab wahi restaurant example.

Tum waiter ko bas bolte ho:

"Ek Paneer Butter Masala."

Bas.

Chef baaki sab khud samajh leta hai.

ORM bhi exactly yehi karta hai.

Tum Python me likhte ho:

```
user.name = "Shivam"
```

ORM usko automatically SQL me convert kar deta hai.

Tum SQL kam likhte ho.

Simple Comparison

Direct SQL	ORM
SQL khud likho	Python objects use karo
Faster	Development faster
More control	Easier maintenance
Beginner ke liye harder	Beginner-friendly

Hamare Project ke liye

Hamare paas hai:

-  Users
-  Organizations
-  Hackathons
-  Teams
-  Submissions
-  Evaluations
-  Certificates

Ye already 15–20 tables ban chuke hain.

Agar hum har jagah manually SQL likhenge, to code bahut bada aur repetitive ho jayega.

Isliye **main recommend karunga ki V1 me ORM use karein.**

Ye industry me bhi bahut common approach hai, especially **FastAPI + SQLAlchemy** ke saath.

Database Access Layer

Backend database se interact karne ke liye ORM use karega.

Reason

- Cleaner code
- Easier maintenance
- Faster development
- Better scalability

★ Golden Rule

Choose the simplest technology that solves today's problem.

Hum duniya ka sabse complex system nahi bana rahe.

Hum ek **reliable, scalable aur maintainable V1** bana rahe hain.

Chapter 39: MVP (Minimum Viable Product)

MVP = The smallest version of the product that solves the main problem.

Hamara main problem kya tha?

"Hackathon ko end-to-end online conduct karna."

Bas.

AI, badges, analytics, themes, dark mode...

Ye sab baad me aa sakta hai.

Step 1: Must Have Features

Ye features **V1 ke bina project incomplete hai.**

Authentication

- User Signup
- User Login
- Forgot Password
- Profile

Organizations

- Create Organization
- Organization Profile

- Invite Organizers
- Transfer Ownership

Hackathons

- Create Draft
- Edit Draft
- Publish
- Public Listing
- Hackathon Details

Teams

- Create Team
- Join Team
- Invite Members
- Team Dashboard

Submission

- Upload Files
- GitHub Link
- Demo Link
- Final Submit

Judging

- Add Judges
- Assign Teams
- Score Project
- Feedback

Results

- Leaderboard
- Winner Announcement

Certificates

- Participation Certificate
- Winner Certificate
- Judge Certificate

Step 2: Good to Have

Ye V1 ke baad aa sakte hain.

- Notifications

- Email reminders
- Search filters
- Dashboard charts
- Hackathon templates
- QR verification

Step 3: Future Features

Ye V2 ya V3 ke liye.

- AI Judge Assistant
- AI Feedback
- Live Chat
- Discussion Forum
- Sponsor Portal
- Payment Gateway
- Public API
- Mobile App
- Multi-language
- Calendar Integration

Step 4: NOT NOW

Ye intentionally V1 me nahi banayenge.

- Blockchain Certificates
- NFT Badges 
- AR/VR
- Complex AI recommendation systems
- Video conferencing
- Built-in IDE
- Real-time collaborative coding

Priority Matrix

Priority	Build Now?
Must Have	 Yes
Good to Have	 Later
Future	 Future
Fancy Ideas	 No

The 80/20 Rule

20% features will deliver 80% of the product's value.

Isliye hum pehle woh 20% banayenge.

Final V1 Scope

Agar koi tumse puche:

"Tumhara platform V1 me kya karega?"

Tum confidently bol sakte ho:

"Ek organization online hackathon create karegi, participants register karenge, teams banayenge, projects submit karenge, judges evaluate karenge, results publish honge aur certificates generate honge."

Agar ye sab smoothly kaam kar gaya...



Tumhara V1 successful hai.

ek important baat.

Ab mujhe tumse ek decision lena hai.

Hum planning phase ke almost end par hain.

Ab coding start karne se pehle do options hain:

Option A (Recommended) ★

Pehle **UI/UX Design** banate hain.

- Participant screens
- Organizer screens
- Judge screens

Figma ya simple wireframes me.

Fayda: Coding ke time confusion zero.

Option B

Seedha backend coding start kar dete hain.

Phir frontend.

Main kya recommend karunga?

100% Option A.

Kyun?

Google, Microsoft, startups... koi bhi directly coding start nahi karta. Pehle wireframes aur screen flow bante hain.

Aur ek aur reason...

Hum itni mehnat se architecture bana chuke hain. Agar ab 2–3 din UI planning me laga diye, to coding 3× faster hogi.



Phase 3: UI/UX & Product Design

Chapter 40: User Journey

User Journey is the complete path a user follows to achieve a goal.

Ab hum har role ki journey banayenge.

1. Participant Journey

Landing Page



Explore Hackathons



View Hackathon Details



Login / Signup



Register



Create / Join Team



Dashboard



Submit Project



Track Evaluation



Results



Certificate

Notice karo... Ye story jaisa flow hai.

Participant ko kabhi randomly page par nahi bhejna chahiye.

2. Organizer Journey

Login



Create Organization



Organizer Dashboard



Create Draft Hackathon



Configure



Publish



Monitor Registrations



Assign Judges



Publish Results



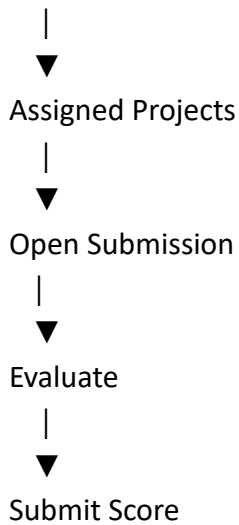
Generate Certificates

3. Judge Journey

Login

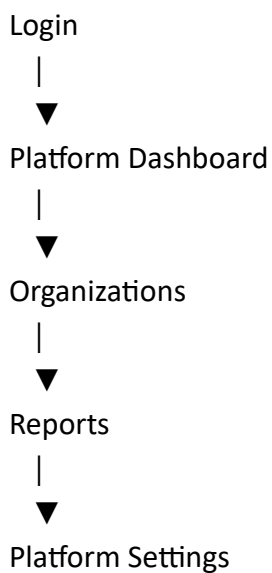


Judge Dashboard



Simple. Judge ko unnecessary screens nahi milni chahiye.

4. Super Admin Journey



★ Golden Rule

Every screen should help the user move one step closer to their goal.

Chapter 41: Design System

A Design System is a collection of reusable design rules and components.

Simple language me:

Ek baar design karo, baar-baar use karo.

1. Design Philosophy

Sabse pehle hum decide karte hain ki hamara platform kaisa feel hona chahiye.

Main 4 options deta hoon.

A. Corporate

Example:

- LinkedIn
- Microsoft

Feel:

- Formal
- Professional

B. Modern Minimal ★

Example:

- Notion
- Linear
- GitHub
- Vercel

Feel:

- Clean
- Spacious
- Fast
- Professional

C. Gamer

Example:

- Discord

Feel:

- Dark
- Colorful
- Animated

D. Fancy

Example:

- Bahut gradients
- Glassmorphism
- Heavy animations

Main Recommendation

Option B - Modern Minimal.

Reason:

Hamara target audience hai:

- Students
- Colleges
- Judges
- Organizations

Sabko simple interface pasand aata hai.

2. Color Philosophy

Yahan ek bahut common mistake hoti hai.

Students sochte hain:

"Jitne zyada colors, utna sundar UI."

Reality:

Professional products 4–6 primary colors hi use karte hain.

Hamare rules:

-  Primary
-  Background
-  Text
-  Success
-  Warning
-  Error

Bas.

3. Typography

Maximum two font families.

Recommended:

Heading



Bold

Body



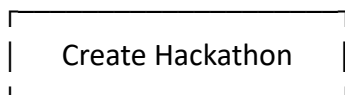
Regular

Har page par alag font nahi.

4. Buttons

Ek hi button style.

Example:



Nahi ki har page par alag design.

5. Cards

Almost sab kuch cards me dikh sakta hai.

Example:

TechSprint 2026
Registration Open
120 Teams
[View Details]

Ek card design. Har jagah reuse.

6. Forms

Forms sabse boring part hote hain.

Isliye:

Har form ka same structure.

Example:

Hackathon Name

Description

Theme

[Dropdown]

[Create]

7. Spacing

Professional UI ka secret.

Consistency is more important than creativity.

Har button ke beech same spacing.

Har card ke andar same padding.

Har heading ka same margin.

Ye chhoti cheez lagti hai...

Lekin isi se UI premium lagta hai.

8. Icons

Ek hi icon library.

Mixed icons nahi.

9. Responsive Design

Hamara platform chalega:

 Desktop

 Mobile

 Tablet

Lekin...

Important Decision ★

V1 me hum **Desktop First** design karenge.

Kyun?

Hackathon organizers aur judges mostly laptop use karenge.

Participants bhi project submission laptop se hi zyada karenge.

Baad me Mobile optimize karenge.

Design Rules

Rule 1

One Primary Button per page.

Example:

Create Hackathon

Baaki buttons secondary.

Rule 2

White space is not empty space.

White space readability improve karta hai.

Rule 3

Never make users think.

User ko manually guess nahi karna chahiye.

Rule 4

Same action = Same design.

Delete button har page par same.

Save button har page par same.

Cancel same.

Chapter 42: Wireframing

A wireframe is a blueprint of a screen without focusing on colors or beauty.

Wireframe ka purpose hai:

- Layout decide karna
- User flow decide karna
- Buttons ki position
- Navigation
- Information hierarchy

Abhi design sundar banana goal nahi hai.

Step 1: Screen List

Sabse pehle poore platform ki screens list banate hain.

Public Screens

Landing Page

Explore Hackathons

Hackathon Details

Login

Signup

Forgot Password

About

Contact

Participant Screens

Dashboard

My Hackathons

My Teams

Team Details

Project Submission

Certificates

Profile

Notifications

Settings

Organizer Screens

Dashboard
Organization
Members
Create Hackathon
Drafts
Published Hackathons
Hackathon Analytics
Judges
Results
Certificates

Judge Screens

Dashboard
Assigned Projects
Evaluation Page
Completed Evaluations
Profile

Super Admin

Dashboard
Users
Organizations
Hackathons
Reports
Audit Logs
Platform Settings

Total Screens

Approximately:

35–45 screens

 Darna mat.

Ye normal hai.

Instagram me 100+ screens hain.

LinkedIn me 200+.

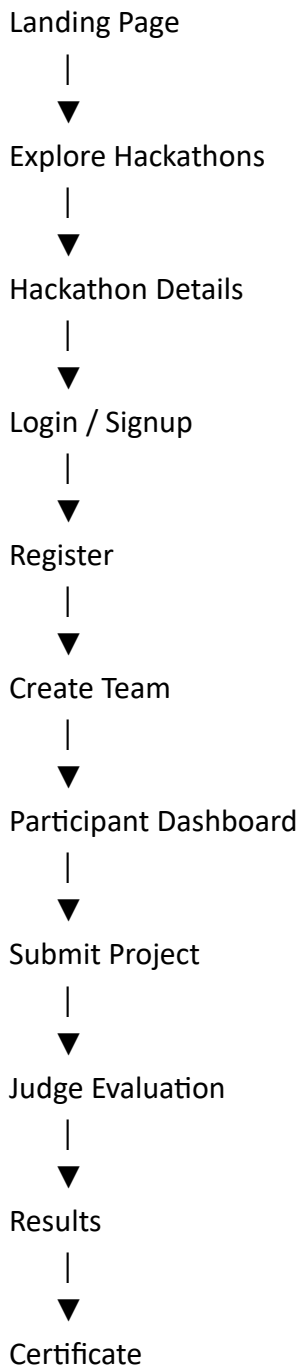
Hum sirf MVP bana rahe hain.

Ab ek bahut important rule

Hum sab screens se start nahi karenge.

Professional teams **Core Flow** se start karti hain.

Hamara Core Flow



Ye hamare product ki backbone hai.

Agar ye flow perfect hai...

To product successful hai.

Screen Design Order

Hum isi order me design karenge:

Phase 1

✓ Landing Page

Phase 2

✓ Explore Hackathons

Phase 3

✓ Hackathon Details

Phase 4

✓ Login

Signup

Forgot Password

Phase 5

Participant Dashboard

Phase 6

Organization Dashboard

Phase 7

Judge Dashboard

Phase 8

Super Admin

Golden Rule

Don't design pages. Design user journeys.

Ye line bahut important hai.

Professional designers page nahi banate.

Wo **experience** design karte hain.

Screen 1: Landing Page (Information Architecture)

The landing page has only one job: convince the visitor to take the next step.

✗ Landing page ka kaam saari information dena nahi hai.

✓ Landing page ka kaam hai user ko explore ya signup tak pahunchana.

Pehle ek question.

Jab koi pehli baar hamari website open karega...

Uske mind me 5 questions honge:

1. Ye platform kya hai?
2. Ye kis ke liye hai?
3. Main yahan kya kar sakta hoon?
4. Kya ye trustworthy hai?
5. Main ab kya karun?

Agar landing page ye 5 questions answer kar deta hai...

Landing page successful hai.

Landing Page Structure

1. Navigation Bar
2. Hero Section
3. Platform Features
4. How It Works
5. Featured Hackathons
6. Why Choose Us
7. Testimonials (Future)
8. FAQ
9. Footer

Notice karo... Ye random order nahi hai.

Ye **psychology** ke according hai.

Section 1: Navigation Bar

LOGO

Explore

Organizations

About

Login

Sign Up

Rule

Navigation me 15 menu items nahi.

Maximum 5–6.

Section 2: Hero Section

Ye poori website ka **heart** hai.

Isme 5 cheezein honi chahiye.

Headline

Short Description

Primary Button

Secondary Button

Hero Illustration

Example layout:

Run & Participate in Hackathons Easily

Everything you need to organize,
join and evaluate hackathons online.

[Explore Hackathons]

[Create Organization]

Illustration

Hero Section ka Psychology

Visitor ke paas sirf **5–8 seconds** hote hain.

Usse instantly samajh aa jana chahiye: "Ye platform mere liye hai."

Primary CTA

Ek page.

Ek primary button.

Example:

 Explore Hackathons

Baaki sab secondary.

Section 3: Platform Features

Yahan hum features list nahi karenge.

Hum **benefits** batayenge.

Example:

✓ Easy Registration

✓ Team Management

✓ Online Submission

✓ Transparent Judging

✓ Automatic Certificates

Notice:

User feature nahi kharidta.

User benefit chahta hai.

Section 4: How It Works

Create



Register



Build



Submit



Evaluate



Win

6 steps.

Section 5: Featured Hackathons

Yahan latest public hackathons.

Hackathon Card

Banner

Title

Registration Open

Prize Pool

Register

Section 6: Why Choose Us

Comparison.

Without attacking anyone.

Example:

Traditional



Manual Registration



Email Submission



Excel Management

Our Platform



Online Registration



Team Management



Online Evaluation



Certificates

Section 7

Testimonials

Abhi nahi.

Future me.

Section 8

FAQ

Example:

How do I participate?

How do I create an organization?

Is it free?

Section 9

Footer

Logo

Quick Links

Privacy

Terms

Contact

Social Media

Chapter 43: Final Tech Stack

Ye woh technologies hain jo **V1 ke liye** use karenge.

Frontend

React + Next.js + TypeScript

Reason:

- Fast
- SEO support
- Industry standard
- Easy deployment

Styling

Tailwind CSS + shadcn/ui

Reason:

- Clean UI
- Fast development
- Professional components

Backend

Python + FastAPI

Reason:

- Fast
- Modern
- Excellent documentation
- AI integration future me easy

Database

PostgreSQL

Reason:

- Reliable
- Scalable
- Industry standard

- Perfect for relational data

ORM

SQLAlchemy

Reason:

- Clean code
- Easy database management

Authentication

- Email/Password
- Google Login
- JWT Authentication

File Storage

Initially:

Local Storage

Future:

Cloud Storage (AWS S3 / Cloudflare R2)

Deployment

Frontend → Vercel

Backend → Railway

Database → PostgreSQL Cloud

Version Control

Git + GitHub

Chapter 44: Folder Structure

hackathon-platform/

├── frontend/

|

├── backend/

|

├── docs/

|

├── database/

|

├── assets/

|

├── .gitignore

|

└── README.md

|
└─ docker-compose.yml (Future)

Chapter 45: Development Roadmap

Phase 1

GitHub Repository

Phase 2

Frontend Setup

Next.js

Tailwind

shadcn/ui

Phase 3

Backend Setup

FastAPI

PostgreSQL

SQLAlchemy

Phase 4

Authentication

Signup

Login

JWT

Phase 5

Organization Module

Phase 6

Hackathon Module

Phase 7

Team Module

Phase 8

Submission Module

Phase 9

Judge Module

Phase 10

Results

Certificates

Phase 11

Testing

Phase 12

Deployment